

JsgSmile MSME Marketplace

Procurement / RFQ / Tender / Bid Management Module

Detailed Functional, Technical, Security and UI Documentation

Prepared for: JsgSmile MSME Marketplace / PugArch Technology Private Limited

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Important branding rule

This document uses public government e-procurement and GeM process references only for process inspiration. Do not copy GeM name, logo, UI assets, text, screenshots, colors, or branding into JsgSmile. The portal name must remain JsgSmile.

Document control item	Details
Project	JsgSmile MSME Marketplace procurement and bid module
Prepared for	Backend completion, frontend participation-page improvement, secure bidding workflow and project documentation upload
Primary objective	Convert static bid UI into secure, login-protected, document-driven, government-style procurement workflow
Scope	Direct purchase, RFQ/tender creation, seller bidding, technical evaluation, financial evaluation, L1 ranking, award and post-award handover
Technology assumption	Existing portal frontend + backend should be inspected first; reuse current auth, user roles, database and file upload conventions

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1. Executive Summary and Implementation Goal

The current procurement/bid module UI in JsgSmile has static API-ready data. The next step is to convert it into a complete production workflow with backend persistence, role-based access, secure document uploads, sealed financial quote handling, technical evaluation, financial evaluation, L1/L2/L3 ranking, award approval and audit trail.

Core rule

Bidding must not be only a frontend form. It must be a secure workflow with login, seller verification, document locking, sealed financial quote, evaluation stages, private seller remarks, award approval and audit logs.

- Only logged-in and verified sellers/vendors can participate in RFQ/tender bids.
- Public users may view limited active bid information, but cannot upload documents, submit quotes or see restricted information.
- Financial quotes must stay sealed/private until the financial evaluation stage.
- Only technically qualified sellers should move into financial evaluation unless a pre-defined alternate method is selected before bid publication.
- Buyer remarks for qualification/disqualification must be private to the respective seller, buyer/evaluator and admin.
- The buyer can recommend award; admin/competent authority can approve final award. By default, the award should go to L1 unless a controlled exception process is configured.

Implementation outcomes required

Area	Required outcome
Backend	Create real database persistence for bids, documents, participation, clarifications, evaluations, awards and audit logs.
Authentication	Protect seller, buyer and admin flows. Redirect unauthenticated participation attempts to login.
Frontend	Replace dummy data with API calls and improve participate page with real forms, drag-and-drop upload, progress, validation and status logic.
Security	File validation, private document storage, signed URLs, quote masking, role authorization and immutable audit logs.
Operations	Clear status transitions from draft to award, plus reports and admin monitoring.

2. Official Process References Used for Inspiration

The JsgSmile flow should be inspired by public government e-procurement patterns, but the implementation must be original and branded only as JsgSmile. The following process ideas are drawn from official public sources listed in the References section.

Reference idea	How it applies to JsgSmile	Source tags
Complete e-procurement lifecycle	Tender publication, online bid submission, bid opening, evaluation result upload and award publication can be represented as a stage-based workflow.	[S1]
GeM bid listing filters and bid types	Use filters such as ongoing bids, bid/RA status, product/service bid, BOQ bid, rate contract, global tender and status filters.	[S2]
Two-stage bid awarding	Technical evaluation should be completed before financial opening; award is linked to financial ranking.	[S3]
Financial document confidentiality	Financial documents should be visible to buyer during financial evaluation only.	[S4]
Clarification response window	Buyer can seek seller clarifications during evaluation, and seller responds within declared time.	[S5]
Two-cover / two-part tendering	Open technical bid first; open financial bid only for techno-commercially successful bidders.	[S6]
L1 and bid splitting controls	Default award to L1. Split award/non-L1	[S3], [S7]

selection must be pre-defined or controlled with approval and audit.
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3. Procurement Types Supported by JsgSmile

3.1 Direct Purchase

Direct purchase is used when the buyer can procure from a catalogue/product/service listing or a direct quotation without running a full competitive bidding lifecycle. It is suitable for smaller, standard or urgent purchases where technical comparison is not required.

Direct purchase stage	Description	Documents
Search and select	Buyer searches product/service and selects seller/catalogue item.	Product catalogue, seller verification profile
Cart / quote confirmation	Buyer enters quantity, delivery location, billing and tax details.	Purchase request, GST details, delivery location
Order creation	Purchase order is generated after buyer confirmation and required approval.	PO, terms, delivery schedule
Delivery and GRN	Seller delivers goods/service; buyer confirms receipt/service completion.	Delivery challan, GRN/service completion certificate
Invoice and payment	Invoice is generated and payment flows through portal settlement setup.	Invoice, payment receipt, settlement record

3.2 RFQ / Tender / Bid-Based Purchase

RFQ/tender purchase is used when multiple sellers/vendors must submit technical and financial offers against a buyer requirement. This is the main process that requires bid documents, seller participation, technical evaluation, financial evaluation, ranking and award.

When to use RFQ/Tender	Reason
Bulk procurement	Multiple sellers should compete on price and capability.
Custom product/service	Buyer needs specification documents, scope of work and compliance sheets.
High value procurement	Transparent evaluation and approvals are needed.
Technical complexity	Seller technical capability must be checked before opening price.
Buyer wants quotations	Multiple quotations create price discovery and ranking.

3.3 Procurement methods to support

Method	Description	Recommended status in JsgSmile
Direct Purchase	Catalogue or direct quote based purchase without bid evaluation.	Phase 1 / active
RFQ	Buyer requests quotations from eligible sellers; technical and financial comparison may be used.	Phase 1 / active
Open Tender	All eligible verified sellers can participate.	Phase 1 / active
Limited Tender	Buyer invites selected verified sellers only.	Phase 2
BOQ Bid	Buyer uploads BOQ sheet and sellers submit price schedule.	Phase 2
Rate Contract	Buyer creates repeat supply terms/rate agreement.	Phase 2
Reverse Auction	Sellers compete by lowering prices after technical qualification.	Phase 3
Service Bid	Tender for service work with scope and manpower/service deliverables.	Phase 1 / active

4. User Roles and Responsibilities

4.1 Role definitions

Role	Primary responsibility	Access level
Requirement User / Department User	Raises internal need for goods/services and uploads requirement details.	Buyer organization internal role
Buyer / Procurement Officer	Converts requirement into RFQ/tender, publishes after approval, evaluates bids and recommends award.	Buyer dashboard
Buyer Approver / Competent Authority	Approves requirement, estimated value, tender document, corrigendum and final award.	Approval dashboard
Seller / Vendor / Supplier	Participates in bids, uploads technical documents and financial quote, responds to clarification and accepts award.	Seller dashboard
Technical Evaluator	Checks technical documents, eligibility and specification compliance; qualifies/disqualifies seller.	Evaluation dashboard
Financial Evaluator	Opens financial quotes of technically qualified sellers and reviews L1/L2/L3 ranking.	Evaluation dashboard
Admin / Platform Authority	Approves bids, monitors process, manages audit logs and final award approval.	Admin dashboard

4.2 RACI matrix

Activity	Requirement user	Buyer	Approver	Seller	Evaluator	Admin
Raise requirement	R	A	C	-	-	-
Prepare RFQ/tender	C	R	A	-	C	C
Approve bid publication	-	R	A	-	-	A/R if platform approval required
Participate in bid	-	-	-	R	-	-
Upload technical documents	-	-	-	R	-	-
Upload sealed financial quote	-	-	-	R	-	-
Clarification request	-	R	-	C/R response	R	C
Technical evaluation	-	A	C	C	R	C
Financial evaluation	-	A	C	-	R	C
Award approval	-	R	A	C	C	A/R if platform approval required

Legend: R = Responsible, A = Accountable, C = Consulted, - = no action.

5. End-to-End RFQ / Tender Lifecycle

Master lifecycle

Requirement Raised -> Requirement Verified -> Tender/RFQ Created -> Submitted for Approval -> Approved and Published -> Seller Participation -> Bid Closed -> Technical Evaluation -> Financial Evaluation -> L1 Ranking -> Award Recommendation -> Admin/Authority Approval -> PO/Work Order -> Delivery/GRN -> Invoice/Payment/Settlement

Stage	Owner	System behavior	Output
1. Requirement raised	Requirement user/buyer	Capture product/service need, quantity, location, timeline, budget and supporting documents.	Requirement record
2. Verification	Buyer/approver/admin	Check completeness, budget approval, category, eligibility and procurement method.	Verified requirement
3. RFQ/tender creation	Buyer	Add bid details, dates, required seller documents, evaluation method and upload tender documents.	Draft bid
4. Approval	Approver/admin	Approve or reject bid before publication.	Approved/open bid
5. Seller participation	Seller/vendor	Login, confirm eligibility, upload technical documents and sealed financial quote, final submit.	Participation record
6. Closing	System	Stop new participation and lock submissions after bid end date.	Closed bid
7. Technical evaluation	Buyer/evaluator	Review documents, ask clarifications, mark qualified/disqualified with private remarks.	Technical result
8. Financial evaluation	Buyer/evaluator	Open quotes only for technically qualified sellers, calculate total evaluated price and ranking.	L1/L2/L3 ranking
9. Award	Buyer/admin/authority	Recommend and approve award, generate award letter and PO/work order.	Awarded bid / PO
10. Post-award	Buyer/seller/payment team	Delivery, GRN, invoice, payment, settlement and closure.	Completed order

6. Buyer Requirement Raising and Verification

6.1 Who raises the requirement?

The requirement can be raised by a buyer organization user, department user, plant/office department, procurement officer or admin-assisted buyer. For example, maintenance department may need industrial safety equipment, IT department may need software services, or HR department may need manpower services.

6.2 Requirement form fields

Field group	Fields
Basic details	Requirement title, category, sub-category, product/service type, urgency, estimated budget, procurement method suggestion
Requirement details	Detailed description, quantity, unit, location, delivery timeline, inspection requirement, warranty requirement, installation requirement
Technical details	Specification text, drawing/design file, quality standard, make/brand preference if justified, compliance requirement
Finance details	Estimated value, budget approval reference, payment terms, tax applicability, funding department
Approvals	Department approver, finance approver, procurement approver, competent authority approval
Attachments	Requirement note, scope of work, technical specification, BOQ/quantity sheet, delivery schedule, internal approval note

6.3 Verification checklist before tender creation

25. Requirement title and description are clear enough for sellers to understand the need.
26. Quantity, unit and delivery location are correct.

27. Estimated value and budget approval are available where required.
28. Technical specifications are not vague or unfairly restrictive.
29. Eligibility criteria are linked to the work requirement and are not arbitrary.
30. Mandatory documents are correctly selected.
31. Procurement method is selected correctly: direct purchase, RFQ, tender, BOQ, rate contract or reverse auction.
32. Dates provide enough time for sellers to prepare bids and respond to clarifications.
33. Terms, delivery schedule and payment terms are clear.

7. RFQ / Tender Creation and Buyer Documents

7.1 Buyer RFQ/tender creation form

Section	Fields required
Basic bid information	Bid title, bid reference number, buyer organization, buyer type, department, contact person, email, mobile, category, procurement method, estimated value, bid validity
Item/service information	Item/service name, description, quantity, unit, technical specification, warranty, AMC/CMC, installation, testing, consignee/delivery location, delivery timeline
Eligibility	GST, PAN, Udyam, experience, turnover, past work orders, certificates, OEM authorization, blacklisting declaration, MSE/MII preferences
Dates	Publish date, start date, pre-bid meeting date, clarification last date, submission end date, technical opening date, financial opening date, expected award date
Commercial terms	EMD required/exempted, EMD amount, document fee, payment terms, taxes, bid validity, price validity, penalty/LD terms
Evaluation method	Two-cover technical + financial, single packet, QCBS/quality-cost method if applicable, L1 default, split award setting, reverse auction setting
Documents required from seller	Document checklist with mandatory/optional flags, accepted file types and templates.

7.2 Buyer documents required for RFQ/tender

Document	Purpose	Mandatory?
Requirement document	Explains business need and scope.	Yes
Technical specification document	Defines exact technical requirements/specifications.	Yes for RFQ/tender
Scope of work	Defines work activities, deliverables and responsibilities.	Yes for services/custom work
Quantity/BOQ sheet	Defines quantities and price schedule format.	Yes if itemized pricing is needed
Terms and conditions document	Defines contract, delivery, payment, penalty and compliance terms.	Yes
Eligibility criteria document	Defines qualification requirements and document proofs.	Yes
Financial quote format	Standardizes seller price submission.	Yes
Drawing/design file	Supports technical design or product detail.	Conditional
Approval note	Proof that buyer organization approved procurement.	Recommended / required by policy
Corrigendum document	Uploaded when bid is modified after publication.	Conditional

7.3 Bid publication flow

34. Buyer saves bid as draft.
35. Buyer submits bid for approval.
36. Approver/admin reviews details and documents.
37. If incomplete, bid is rejected with reason and returned to buyer.
38. If approved, bid becomes scheduled/open depending on start date.

39. System notifies matching sellers based on category, location, keywords and verification status.

40. Bid appears on latest bids, bid listing, seller dashboard and buyer dashboard.

8. Seller Notification, Login Gating and Eligibility

8.1 Seller notification logic

Matching parameter	How it should work
Category	Notify sellers registered in matching category/sub-category.
Location	Notify sellers servicing buyer delivery district/state or all-India sellers.
Keywords	Match item/service keywords with seller profile/catalogue.
Verification status	Prioritize verified sellers.
Past performance	Future enhancement: notify sellers with relevant historical orders or work categories.
MSME/Udyam status	Notify MSME sellers if bid has MSME preference or buyer targets MSME procurement.

8.2 Login and eligibility rules

User state	Participation behavior	Message
Not logged in	Redirect to login before participation.	Please login as a verified seller/vendor to participate in this bid.
Logged in as buyer	Block participation.	Only verified sellers/vendors can participate in bids.
Logged in as admin	Block seller participation unless impersonation/support mode exists.	Admins can monitor bids but cannot participate as sellers.
Logged in seller but profile incomplete	Block participation.	Please complete seller verification before participating.
Verified seller, bid open	Allow start participation.	You can participate in this bid.
Verified seller, bid closed	Disable participate action.	Bid submission date has ended.
Seller already participated	Show existing participation status instead of creating duplicate.	You have already participated in this bid.

9. Seller Bid Participation Page and Required UI

The current participate page must be converted from static placeholders into a real stage-based submission form. The left submission stepper should update the right-side form according to the selected step. Each step must save draft data and validate required fields before allowing final submission.

9.1 Top participation header

Header element	Required behavior
Bid title and number	Show bid title, bid number and buyer organization.
Status badges	Show Open/Closing Soon/Closed, participation draft/submitted, verification status.
Closing countdown	Show days/hours remaining. Red/yellow warning near deadline.
Login/verification alert	Show warning if user is not logged in or seller is not verified.
Document download buttons	Download tender document, specification, BOQ format and corrigendum if applicable.

9.2 Left stepper structure

Step	Purpose	Right-side form/content
1. View bid	Seller reviews bid information.	Bid summary, dates, buyer info, delivery location, eligibility criteria, required docs and document downloads.
2. Check eligibility	Seller confirms and system checks profile eligibility.	Checklist, auto-check GST/PAN/Udyam/profile/bank verification, declaration checkboxes.
3. Technical offer	Seller enters offered item/service details.	Make/brand, model, offered item

		description, delivery timeline, warranty, compliance summary.
4. Upload technical documents	Seller uploads mandatory/optional technical documents.	Drag-and-drop upload components, file validation, progress, preview/remove/reupload.
5. Financial quote	Seller enters quote and uploads sealed financial documents.	Base amount, GST, freight, installation, discount, total, quote PDF, BOQ Excel/PDF.
6. Review and declaration	Seller reviews all data before final submit.	Summary of eligibility, technical offer, uploads, quote total and declaration checkboxes.
7. Submit bid	Seller confirms final submission.	OTP/email/mobile verification, final submit button, confirmation modal and receipt.
8. Track status	Seller sees lifecycle progress.	Bid published, participated, technical evaluation, financial evaluation, L1 selection, awarded/not selected.

9.3 Real upload component requirements

Upload feature	Requirement
Drag and drop	Allow user to drag files into upload zone.
Choose file button	Provide visible button for normal file selection.
Required/optional badge	Show mandatory documents clearly.
Accepted formats	Display allowed file formats: PDF, DOC, DOCX, XLS, XLSX, PNG, JPG, JPEG.
Max size note	Display maximum file size and validate server-side.
Upload progress	Show progress bar or uploading state.
Preview	Show uploaded file name, size, type and view/download action if allowed.
Remove/re-upload	Allow until final submission or bid closure.
Error state	Show file type/size/virus scan/server errors clearly.
Lock after submit	After final submission, documents cannot be deleted or replaced.

10. Technical Document Upload Requirements

10.1 Seller technical documents

Document category	Description	Mandatory logic
GST certificate	Proof of GST registration.	Mandatory if buyer selected GST requirement.
PAN	Business/person PAN proof.	Mandatory for sellers.
Udyam certificate	MSME proof for preference/eligibility.	Mandatory if seller claims MSME benefit or buyer requires it.
Company registration / Shop Act / incorporation proof	Legal proof of business.	Mandatory depending on seller type.
Experience certificate / work order	Proof of similar work/supply.	Conditional based on buyer criteria.
Completion certificate	Proof of completed previous work.	Conditional.
Product catalogue / service profile	Shows offered product/service capability.	Recommended/conditional.
Technical compliance sheet	Seller maps each technical requirement to compliance response.	Mandatory for technical bids.
Quality certificate	BIS/ISI/ISO or category-specific certificate.	Conditional.
OEM authorization letter	Required if seller is not manufacturer and item needs authorization.	Conditional.
Make in India/local content declaration	Local content/supporting declaration if applicable.	Conditional.
EMD proof or exemption proof	Proof of EMD payment or exemption.	Conditional.
Undertaking/declaration	Non-blacklisting, acceptance of terms, authenticity of documents.	Mandatory for formal tender.

10.2 Technical offer fields

Field	Purpose
Offered item/service name	Seller identifies exactly what is being offered.

Make/brand	Product make/brand if applicable.
Model	Model or version information.
Technical compliance response	Compliant / partially compliant / not applicable with remarks.
Delivery timeline	Seller proposed delivery timeline.
Warranty/AMC/CMC	Warranty and support obligations.
Deviation statement	Any deviation from tender specification.
Remarks	Seller clarification or technical explanation.

11. Financial Quote Upload and Sealed Bid Rules

Financial quote data must be treated as sealed confidential information. It should not be visible to buyer/evaluator during technical evaluation. It becomes available only after technical evaluation is completed and the buyer/evaluator opens financial evaluation for technically qualified sellers.

11.1 Financial quote fields

Field	Description
Base quoted amount	Base price excluding taxes/charges.
GST percentage and amount	Applicable GST calculation.
Freight/transport charges	Charges for delivery/logistics.
Installation/commissioning charges	Installation cost if applicable.
AMC/CMC/warranty cost	Service/support cost if included in evaluated price.
Discount	Commercial discount given by seller.
Total quoted amount	Auto-calculated total amount.
Total evaluated amount	System calculated amount used for ranking; includes loading factors if configured.
Delivery timeline	Seller commercial delivery commitment.
Price validity period	How long the quote remains valid.
Financial quote PDF	Seller official signed quote.
BOQ price sheet	Excel/PDF price schedule where BOQ method is used.

11.2 Sealed quote access rules

Stage	Who can see financial quote?	Rule
Before submission	Seller owner only	Draft data visible to the seller who created it.
After seller final submit but before bid close	Seller owner only; buyer sees participation count only	Quote locked and sealed.
Technical evaluation stage	Seller owner only; buyer/evaluator cannot open financial quote	Financial data remains masked.
Financial evaluation opened	Buyer/evaluator/admin can view qualified sellers financial quotes	Only technically qualified sellers are included.
Public result	Only allowed award/ranking summary if configured	Do not expose confidential details unnecessarily.

12. Clarification Process

12.1 Clarification flow

41. Bid closes and technical evaluation starts.
42. Buyer/evaluator reviews seller technical documents.
43. If missing/unclear, buyer raises clarification for one seller.
44. System notifies only that seller.
45. Seller responds with text and optional documents before due date.
46. Buyer reviews response and marks clarification as completed, reopened, rejected or expired.
47. Clarification record is preserved in audit history and visible only to allowed users.

12.2 Clarification fields

Field	Description
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Clarification request number	Unique clarification sequence number.
Bid number and participation number	Links request to specific bid and seller submission.
Clarification type	Document missing, technical mismatch, certificate validity, compliance response, other.
Question/description	Buyer/evaluator request text.
Due date/time	Deadline for seller response.
Seller response	Seller explanation or correction.
Uploaded response documents	Optional files uploaded by seller.
Status	Pending, responded, completed, reopened, rejected, expired.
Remarks	Private buyer/evaluator remarks.

Privacy rule

Clarification details and remarks for Seller A must not be visible to Seller B or public users. Buyer/evaluator/admin can see all; each seller can see only their own clarification.

13. Technical Evaluation Process

Technical evaluation starts only after the bid submission deadline. The buyer/evaluator verifies seller documents, eligibility, technical compliance and clarification responses before marking a seller as qualified or disqualified.

13.1 Technical evaluation checklist

Checklist item	Evaluation question
GST/PAN/Udyam	Are required statutory IDs valid and matching seller profile?
Experience	Does seller satisfy years of experience and similar work/supply requirements?
Turnover	Does seller satisfy turnover/financial capability requirement?
Certificates	Are BIS/ISI/ISO/OEM/quality certificates uploaded and valid?
Technical compliance	Does offered item/service meet required specifications?
Delivery capability	Can seller deliver to location within timeline?
EMD/exemption	Is EMD proof or exemption proof valid if required?
Clarifications	Were requested clarifications answered satisfactorily?
Terms acceptance	Has seller accepted tender terms and declarations?

13.2 Technical status rules

Status	Meaning	Next action
PENDING	No evaluation completed yet.	Evaluator reviews documents.
UNDER_REVIEW	Evaluator is reviewing submission.	Continue checks / ask clarification.
CLARIFICATION_REQUIRED	Buyer needs additional response/document.	Seller responds before due date.
QUALIFIED	Seller passed technical evaluation.	Move to financial evaluation.
DISQUALIFIED	Seller failed criteria.	Do not open financial quote; show private remarks to seller.

13.3 Private remarks visibility

User	Can view
Seller A	Only own technical status and own buyer remarks.
Seller B	Only own technical status and own buyer remarks.
Buyer/evaluator	All sellers technical statuses, documents and remarks for own bid.
Admin	All evaluation records for monitoring and audit.
Public user	No private remarks or seller-specific documents.

14. Financial Evaluation, L1/L2/L3 Ranking and Award

14.1 Financial evaluation flow

48. Technical evaluation must be completed for all sellers.
49. System filters only technically qualified sellers.
50. Buyer/evaluator clicks Open Financial Evaluation.
51. Financial quotes and BOQ price sheets of qualified sellers are unlocked for buyer/evaluator/admin.
52. System calculates total evaluated amount for each qualified seller.
53. System sorts sellers from lowest evaluated amount to highest evaluated amount.
54. System assigns L1, L2, L3, L4, etc. ranking.
55. Buyer reviews price comparison and recommends award.
56. Admin/competent authority approves final award and system creates award record/letter.

14.2 Ranking logic

Rank	Rule
L1	Lowest total evaluated amount among technically qualified sellers.
L2	Second lowest evaluated amount.
L3	Third lowest evaluated amount.
L4 and beyond	Continue sequential ranking by amount.
Tie	Apply configured tie-breaker: earlier submission time, higher technical score, MSME preference or manual authority decision as per published criteria.

14.3 Award rules

Scenario	Recommended JsgSmile behavior
Normal case	Recommend L1 seller for award.
Buyer attempts non-L1 selection	Block by default and require strong reason, competent authority approval and admin approval.
L1 refuses order	Allow next-ranked seller only with reason, approval and audit.
Split award pre-defined	Allow award split only if splitting ratio was configured before bid publication.
Quality-cost based selection pre-defined	Allow final selection based on pre-published method and scoring formula.

14.4 Award outputs

Generated item	Purpose
Award letter	Official record of selected seller and amount.
Purchase order/work order	Commercial order issued to seller.
Contract copy	Terms and obligations after award.
Financial approval note	Internal buyer approval record.
Seller acceptance	Seller acceptance of award/order.
Delivery schedule	Delivery milestones and dates.

15. Direct Purchase Flow

Direct purchase is separate from RFQ/tender. It should be faster and should not require technical/financial evaluation, L ranking or bid award. Use it for catalogue and small/simple purchases.

Direct purchase step	System feature
Buyer searches product/service	Marketplace search and filters.
Buyer reviews seller	Verified seller profile, ratings, documents if available.
Add to cart/request direct quote	Quantity, delivery location, taxes, expected date.
Approval if required	Buyer internal approval based on value/category.
Order confirmation	PO/order generated.
Seller acceptance	Seller accepts/rejects order.

Delivery and GRN	Buyer confirms goods/services received.
Invoice and payment	Payment and settlement flow.

16. Post-Award Order, GRN, Invoice and Settlement

After award, the procurement flow becomes an order execution flow. For JsgSmile, this should connect to the planned portal payment and Bandhan Bank nodal/escrow settlement model.

57. Award is approved and PO/work order is generated.
58. Seller accepts award/order and confirms delivery timeline.
59. Seller delivers goods or completes service.
60. Buyer records GRN/service completion certificate.
61. Invoice is generated or uploaded.
62. Buyer initiates payment through portal.
63. Payment is processed through payment gateway / bank setup.
64. Payment amount is held in nodal/escrow account as per business flow.
65. Settlement engine triggers seller settlement after applicable settlement period.
66. Bank confirmation updates order status to settled.
67. Final receipt/invoice alerts are sent to buyer and seller.

17. Data Models and Status Machine

17.1 Core database models

Model	Purpose	Key fields
ProcurementBid	Stores RFQ/tender master details.	id, bidNumber, title, buyerId, buyerType, category, quantity, estimatedValue, dates, status, approvalStatus, lifecycleStage, terms, eligibility, requiredDocuments
BidDocument	Stores buyer uploaded tender/specification/BOQ documents.	id, bidId, documentType, fileName, fileUrl/fileKey, mimeType, fileSize, visibility, uploadedBy
BidParticipation	Stores seller participation summary and statuses.	id, bidId, sellerId, participationNumber, statuses, quotedAmount, totalAmount, makeBrand, model, submissionStatus, submittedAt, rank
BidParticipationDocument	Stores seller uploaded technical and financial documents.	id, participationId, documentCategory, file metadata, documentStatus, uploadedAt
BidClarification	Stores buyer-seller clarification requests and responses.	id, bidId, participationId, sellerId, buyerId, requestNumber, question, response, status, dueDate
BidClarificationFile	Stores files attached to clarification response/request.	id, clarificationId, file metadata, uploadedBy
BidEvaluation	Stores technical and financial evaluation records.	id, bidId, participationId, evaluationType, status, remarks, score, evaluatorId, evaluatedAt
BidAward	Stores final award record.	id, bidId, participationId, sellerId, awardedAmount, awardStatus, awardedBy, awardedAt, remarks
ProcurementAuditLog	Immutable audit history for important actions.	id, userId, role, entityType, entityId, action, oldValue, newValue, ipAddress, userAgent, createdAt

17.2 Bid status machine

From status	Allowed next status	Trigger
DRAFT	PENDING_ADMIN_APPROVAL	Buyer submits bid for approval.
PENDING_ADMIN_APPROVAL	APPROVED or REJECTED	Admin/authority decision.
APPROVED	OPEN	Start date reached / publish action.
OPEN	CLOSED	End date reached / manual close allowed by policy.

CLOSED	TECHNICAL_EVALUATION	Buyer starts technical evaluation.
TECHNICAL_EVALUATION	TECHNICAL_EVALUATION_COMPLETE	All sellers evaluated.
TECHNICAL_EVALUATION_COMPLETE	FINANCIAL_EVALUATION	Buyer opens financial evaluation.
FINANCIAL_EVALUATION	L1_GENERATED	System generates ranking.
L1_GENERATED	AWARD_RECOMMENDED	Buyer recommends award.
AWARD_RECOMMENDED	AWARDED	Admin/authority approves award.
Any pre-award state	CANCELLED	Authorized cancellation with reason and audit.

17.3 Participation statuses

Status group	Values
Submission status	DRAFT, TECHNICAL_UPLOADED, FINANCIAL_UPLOADED, SUBMITTED, WITHDRAWN, LOCKED
Technical status	PENDING, UNDER_REVIEW, CLARIFICATION_REQUIRED, QUALIFIED, DISQUALIFIED
Financial status	LOCKED, NOT_OPENED, OPENED, EVALUATED
Final status	PENDING, L1, L2, L3, L4_PLUS, AWARDED, NOT_SELECTED, REJECTED

18. API Endpoint Specification

18.1 Public APIs

Endpoint	Method	Purpose	Access
/api/bids	GET	Fetch approved/open bid listing with filters, search, sorting and pagination.	Public limited
/api/bids/:bidId	GET	Fetch public bid details and public documents only.	Public limited
/api/bids/:bidId/documents/public	GET	List public bid documents only.	Public limited

18.2 Seller APIs

Endpoint	Method	Purpose	Rules
/api/bids/:bidId/participate	POST	Start participation.	Seller login + verification + bid open + no duplicate.
/api/bids/:bidId/participation/:id/technical-offer	PUT	Save technical offer details.	Owner seller only; before final submit.
/api/bids/:bidId/participation/:id/technical-documents	POST	Upload technical docs.	Owner seller only; file validation.
/api/bids/:bidId/participation/:id/financial-quote	POST	Save/upload sealed financial quote.	Owner seller only; before close/final submit.
/api/bids/:bidId/participation/:id/submit	POST	Final submit participation.	Validate mandatory docs + quote + OTP; lock records.
/api/seller/bids	GET	List seller participated bids.	Seller sees own statuses only.
/api/seller/bids/:bidId/status	GET	Seller bid tracking timeline.	Seller owner only.
/api/bids/:bidId/clarifications/:clarificationId/respond	POST	Seller responds to clarification.	Seller owner only; before due date.

18.3 Buyer APIs

Endpoint	Method	Purpose	Rules
/api/buyer/bids	POST	Create bid as draft.	Buyer role only.
/api/buyer/bids/:bidId	PUT	Edit draft/rejected bid.	Only owner buyer; only editable states.
/api/buyer/bids/:bidId/submit-	POST	Submit bid for approval.	Validate required fields/docs.

for-approval			
/api/buyer/bids	GET	List buyer own bids.	Buyer role.
/api/buyer/bids/:bidId/ participants	GET	View participants after close/evaluation start.	No financial quote before financial evaluation.
/api/buyer/bids/:bidId/ clarifications	POST	Ask clarification to seller.	Technical evaluation stage only.
/api/buyer/bids/:bidId/technical- evaluation	POST	Evaluate seller technical bid.	Qualified/disqualified + private remarks.
/api/buyer/bids/:bidId/complete- technical-evaluation	POST	Complete technical stage.	All sellers resolved.
/api/buyer/bids/:bidId/open- financial-evaluation	POST	Open financial evaluation and generate ranking.	Only after technical completed.
/api/buyer/bids/:bidId/ recommend-award	POST	Recommend award.	Default L1; non-L1 needs controlled reason/approval.

18.4 Admin APIs

Endpoint	Method	Purpose
/api/admin/bids	GET	List all bids with approval/evaluation status.
/api/admin/bids/:bidId/approve	POST	Approve pending bid.
/api/admin/bids/:bidId/reject	POST	Reject pending bid with reason.
/api/admin/bids/:bidId/audit	GET	View bid audit trail.
/api/admin/bids/:bidId/participants	GET	View all participants and evaluation status.
/api/admin/bids/:bidId/final-award-approval	POST	Approve final award.
/api/admin/procurement/reports	GET	Export procurement reports.

19. Security, Audit and Access Control

19.1 Backend access rules

Area	Rule
Authentication	All participate, upload, buyer, evaluator and admin APIs must require valid session/JWT.
Authorization	Enforce role checks on backend. Frontend checks are not enough.
Seller verification	Seller must complete GST/PAN/Udyam/business/bank verification as applicable before participation.
Ownership checks	Seller can access only own participation/documents/remarks. Buyer can manage only own bids. Admin can monitor all.
State machine	Only allowed status transitions should be accepted.
Time locking	No participation/upload/edit after bid end date or final submission.
Financial masking	Do not expose financial quotes before financial evaluation opening.

19.2 File upload security

Control	Requirement
Allowed file types	PDF, DOC, DOCX, XLS, XLSX, PNG, JPG, JPEG.
Server-side MIME validation	Validate actual MIME/content, not only extension.
File size	Enforce max file size per document and total upload limit.
Sanitize file names	Remove path traversal, script names, unsafe characters.
Private storage	Store technical/financial docs in private bucket/storage.
Signed URLs	Generate short-lived signed/private URLs for authorized users.
Virus scanning hook	Add integration point for malware scanning before accepting files.
Audit	Log upload, delete/reupload before submit and file access events where required.

19.3 Audit log actions

Action category	Examples
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Bid creation	BID_CREATED, BID_SUBMITTED_FOR_APPROVAL, BID_APPROVED, BID_REJECTED, BID_PUBLISHED
Participation	BID_PARTICIPATED, TECHNICAL_DOC_UPLOADED, FINANCIAL_DOC_UPLOADED, PARTICIPATION_SUBMITTED
Clarification	CLARIFICATION_REQUESTED, CLARIFICATION_RESPONDED, CLARIFICATION_COMPLETED
Evaluation	TECHNICAL_EVALUATION_STARTED, SELLER_QUALIFIED, SELLER_DISQUALIFIED, FINANCIAL_EVALUATION_OPENED, L1_GENERATED
Award	AWARD_RECOMMENDED, AWARD_APPROVED, BID_AWARDED, PO_GENERATED
Security	UNAUTHORIZED_ACCESS_ATTEMPT, FILE_ACCESS_DENIED, INVALID_STATUS_TRANSITION

19.4 Standard API response format

Success response:

```
{ "success": true, "message": "Operation successful", "data": { } }
```

Error response:

```
{ "success": false, "message": "Clear user-friendly error message", "errorCode": "ERROR_CODE" }
```

Error code	Meaning
UNAUTHORIZED	User is not logged in.
FORBIDDEN_ROLE	Logged-in user role is not allowed.
SELLER_NOT_VERIFIED	Seller verification is incomplete.
BID_NOT_OPEN	Bid is not currently open for participation.
BID_ALREADY_CLOSED	Bid deadline has passed.
DUPLICATE_PARTICIPATION	Seller already participated in this bid.
PARTICIPATION_ALREADY_SUBMITTED	Participation is locked after final submit.
INVALID_STATUS_TRANSITION	Requested state change is not allowed.
FINANCIAL_NOT_OPENED	Financial quote cannot be viewed yet.
FILE_TYPE_NOT_ALLOWED	Uploaded file type is not accepted.
FILE_SIZE_EXCEEDED	Uploaded file exceeds allowed size.

20. Frontend Page Improvements and UX Requirements

20.1 Participate page redesign

Current issue	Required improvement
Static text inputs for uploads	Replace with real drag-and-drop upload components and choose-file buttons.
Start participation available without login gating	Redirect unauthenticated users to login and block non-seller roles.
Stepper does not act like complete forms	Each step should show relevant form content, save draft and validate before next step.
Financial quote field is too simple	Add proper quote calculator, BOQ/quote upload, sealed quote warning and total amount logic.
Track status is basic	Add government-style lifecycle tracker with current stage, completed stages and remarks where allowed.
No receipt after submit	Generate participation receipt with participation number and timestamp.

20.2 Visual design requirements

- Light theme only using white, navy blue, light blue, saffron/orange accent, green success and light gray backgrounds.
- Use clean government-enterprise cards, status badges, table headers and sticky sidebars.
- Use smooth transitions, hover lift on cards and gentle step animations; avoid flashy effects.
- Use loading skeletons for bid details, upload zones, participants and result pages.
- Show clear empty states: no documents uploaded, no clarifications, no participants, no results yet.

- All pages must be fully responsive for mobile, tablet and desktop.
- Use accessible labels, headings, buttons and keyboard-friendly upload controls.

20.3 Buyer and admin UI pages

Page	Required features
Buyer create bid	Multi-section form, document upload, eligibility criteria builder, required documents checklist, save draft, submit approval.
Buyer participants	Show participants after close/evaluation start, technical docs, clarification, evaluation actions and masked financial data.
Technical evaluation	Seller cards/table, documents, compliance checklist, qualify/disqualify, private remarks and clarification action.
Financial evaluation	Unlock qualified sellers only, price comparison, L ranking table, total evaluated amount and award recommendation.
Admin bid management	Approval/rejection, audit trail, participants, evaluation progress, final award approval and reports.

21. Backend Task Breakdown for Codex

21.1 Implementation sequence

68. Inspect existing auth, role system, database schema, API route pattern, file upload mechanism and frontend API service pattern.
69. Add database migrations/models for procurement bids, documents, participation, clarification, evaluation, award and audit logs.
70. Implement backend constants/enums for statuses, roles, document categories and allowed transitions.
71. Implement public bid listing/detail APIs with filtering, pagination and restricted field selection.
72. Implement buyer bid creation, document upload, draft editing and submit-for-approval APIs.
73. Implement admin approval/rejection APIs and audit log visibility.
74. Implement seller participation start, technical document upload, financial quote upload, final submit and status APIs.
75. Implement clarification request/response APIs with private visibility.
76. Implement technical evaluation APIs with qualify/disqualify, private remarks and evaluation report upload.
77. Implement financial evaluation opening, quote unsealing, price calculation and L ranking generation.
78. Implement award recommendation and admin final award approval.
79. Replace frontend dummy data with live API calls and improve participate page UI.
80. Add tests for public/seller/buyer/admin flows and security boundaries.

21.2 Backend implementation guardrails

- Do not break existing login, marketplace, cart, product listing, buyer, seller, admin or payment functions.
- Use transactions for final submit, evaluation completion, financial opening, ranking and award approval.
- Prevent duplicate participation with database unique constraint on bidId + sellerId.
- Never expose quotedAmount or financial files before financial opening.
- Use signed/private URLs for restricted documents.
- Every important action must create an audit log entry.
- Add validation schemas for dates, amounts, file metadata and required fields.

22. Acceptance Tests and QA Checklist

22.1 Public user tests

81. Open /bids without login and verify active/open bid listing loads.
82. Open bid details without login and verify only public details/documents appear.
83. Click Participate without login and verify redirect to login.
84. Try restricted API directly without login and verify UNAUTHORIZED.

22.2 Seller tests

85. Unverified seller cannot participate and receives SELLER_NOT_VERIFIED.

- 86. Verified seller can start participation only when bid is open.
- 87. Same seller cannot participate twice in same bid.
- 88. Seller can upload technical documents with validation and progress.
- 89. Seller can enter financial quote and upload quote document before deadline.
- 90. Seller cannot edit documents/quote after final submit.
- 91. Seller sees only own technical remarks and clarifications.
- 92. Seller cannot access another seller documents via API.

22.3 Buyer tests

- 93. Buyer can create draft bid and upload tender/specification documents.
- 94. Buyer can submit bid for approval.
- 95. Buyer cannot publish without admin/authority approval if approval is configured.
- 96. Buyer can view participants only after close/evaluation start.
- 97. Buyer cannot view financial quote during technical evaluation.
- 98. Buyer can ask clarification and receive seller response.
- 99. Buyer can qualify/disqualify seller with private remarks.
- 100. Buyer can open financial evaluation only after technical evaluation completed.
- 101. L ranking is generated correctly from lowest to highest evaluated amount.
- 102. Non-L1 recommendation is blocked or requires configured approval reason.

22.4 Admin tests

- 103. Admin can approve/reject bids with reason.
- 104. Admin can view audit logs.
- 105. Admin can view all participants and evaluation progress.
- 106. Admin can approve final award.
- 107. Admin can export reports.

22.5 Security tests

- 108. Public cannot access restricted document URLs.
- 109. Financial quote is hidden until financial evaluation.
- 110. Invalid file type is rejected on server.
- 111. Bid end date blocks participation and document upload.
- 112. Invalid status transition is rejected.
- 113. Audit log records IP/user agent for important actions.

23. Full Implementation Prompt for Codex

Copy the following prompt into Codex/Claude after adding this PDF into the project docs folder.

You are working on the JsgSmile MSME Marketplace. The procurement/bid frontend UI exists with static API-ready data. Convert it into a real secure backend-integrated procurement workflow.

Important: Do not use GeM name, logo, text, screenshots or branding anywhere. Use public government procurement process ideas only as inspiration. Portal branding must remain JsgSmile.

Before coding, inspect the existing authentication, user roles, database schema, API route conventions, file upload system, frontend API service pattern and deployment structure.

Implement two purchase flows: Direct Purchase and RFQ/Tender/Bid-Based Purchase. Direct Purchase should not require bidding. RFQ/Tender should support buyer bid creation, approval, seller participation, technical evaluation, financial evaluation, L1/L2/L3 ranking and award.

Only logged-in and verified sellers/vendors can participate in bids. Public users can view limited open bid details only. If public user clicks Participate, redirect to login. If logged-in user is not a seller/vendor, block participation. If seller is not verified, show seller verification error.

Add/extend database models for ProcurementBid, BidDocument, BidParticipation, BidParticipationDocument, BidClarification, BidClarificationFile, BidEvaluation, BidAward and ProcurementAuditLog. Use strict enums and status transitions.

Implement public APIs: GET /api/bids and GET /api/bids/:bidId with filters, search, sorting and pagination, exposing only public data.

Implement seller APIs for start participation, save technical offer, upload technical documents, upload sealed financial quote, final submit participation, view own bid status and respond to clarification.

Implement buyer APIs for create bid, edit draft/rejected bid, submit for approval, list own bids, view participants after close/evaluation start, ask clarification, technical evaluation, complete technical evaluation, open financial evaluation and recommend award.

Implement admin APIs for approve/reject bids, view audit, view participants, approve final award and export procurement reports.

Financial quote must remain sealed/private until financial evaluation. Buyer/evaluator/admin must not see financial quotes during technical evaluation. Only technically qualified sellers should move to financial evaluation.

Technical evaluation must allow qualify/disqualify with private remarks. Seller can see only own remarks. Buyer/admin can see all remarks for authorized bids.

Clarification must be private to the specific seller and buyer/evaluator/admin. Support question, due date, response, files and statuses: pending, responded, completed, reopened, rejected, expired.

Financial evaluation should calculate total evaluated amount and rank sellers L1, L2, L3, L4 etc from lowest to highest. Default award should be L1. Non-L1 selection should be blocked by default or require reason, competent authority/admin approval and audit log.

Add real file upload UI and backend handling. Allowed files: PDF, DOC, DOCX, XLS, XLSX, PNG, JPG, JPEG. Validate MIME and size server-side. Store restricted files privately. Use signed/private URLs. Sanitize file names and add malware scan hook if available.

Improve participate page UI: top bid header, login/verification alert, status badges, closing countdown, left stepper, real drag-and-drop uploads, financial quote calculator, review/declaration, OTP/email verification style final submit and participation receipt.

Replace static frontend dummy data with live API calls for latest bids, bid listing, bid details, participation, clarification, buyer publish form, admin bid management and results.

Add loading skeletons, empty states, responsive design, accessible labels, and government-enterprise light theme styling.

Add tests for public, seller, buyer, admin and security scenarios. Verify that seller A cannot view seller B documents/remarks and financial quotes are hidden until financial opening.

After implementation, provide a summary of files changed, backend modules added, migrations/models added, APIs added, RBAC protections, file upload security, lifecycle implemented, frontend integration, tests run and pending production tasks.

24. References

The links below are included for engineering research and process inspiration. Do not copy their branding, UI assets or exact text into JsgSmile.

Tag	Source	URL	Use in this document
S1	CPPP General Guidelines for e-Procurement	https://eprocure.gov.in/cpppp/sites/default/files/guidelines/cpppp_guidelines.pdf	Complete tendering process: online publishing, online bid submission, online bid opening, bid evaluation results and award publication.
S2	GeM All Bids public listing	https://bidplus.gem.gov.in/bidresultlists	Bid/RA listing filters and bid types such as product/service bid, custom bid, BOQ bid, rate contract, global tender and status filters.
S3	GeM Bid Awarding and Bid Health Score training PDF	https://assets-bg.gem.gov.in/resources/upload/shared_doc/training_content/Bid-Awarding-and-Bid-Health-Score-1771250319.pdf	Two stages of bid awarding: technical and financial; L1 bypass rules.
S4	GeM Latest page	https://gem.gov.in/latest	Financial documents visible to buyer during financial

			evaluation alone.
S5	GeM Bid Creation Product training PDF	https://assets-bg.gem.gov.in/resources/upload/shared_doc/training_content/Bid-Creation-Product-2-1656482992.pdf	Bid duration, offer validity, clarification duration and bid creation concepts.
S6	CPPP Model Tender Document - Services	https://eprocure.gov.in/cppp/sites/default/files/standard_biddingdocs/MTD%20Srvices%20NIC.pdf	Two-cover tender concept: techno-commercial evaluation first, financial bids only for successful bids.
S7	GeM General Terms and Conditions 4.0	https://assets-bg.gem.gov.in/resources/upload/shared_doc/gtc/GeM-GTC-40-1740735825.pdf	Bid splitting and L1 reasonable price concepts.
S8	GeM L1 Purchase Product training PDF	https://assets-bg.gem.gov.in/resources/upload/shared_doc/training_content/B17-L1-Purchase-Product-1685428523.pdf	Direct/L1 purchase process inspiration.
S9	GeM Buyer FAQs	https://gem.gov.in/userFaqs	Public visibility of evaluation decisions and awarded contracts as government-style transparency reference.

Appendix A: Process Maps

A.1 RFQ/Tender process map

Requirement Raised

- > Requirement Verified
- > Procurement Method Selected
- > Tender/RFQ Draft Created
- > Buyer Documents Uploaded
- > Submitted for Approval
- > Approved and Published
- > Sellers Notified
- > Seller Login + Verification
- > Seller Technical Documents Uploaded
- > Seller Sealed Financial Quote Uploaded
- > Seller Final Submit
- > Bid Closed
- > Technical Evaluation
- > Clarification if Required
- > Qualified/Disqualified with Private Remarks
- > Financial Evaluation for Qualified Sellers

- > L1/L2/L3 Ranking
- > Award Recommendation
- > Admin/Authority Approval
- > Award Letter / PO
- > Delivery / GRN / Invoice / Payment / Settlement

A.2 Direct purchase process map

Buyer searches catalogue/service

- > Selects product/seller
- > Adds quantity and delivery location
- > Confirms price/taxes
- > Internal approval if required
- > Order/PO generated
- > Seller accepts
- > Delivery/service completion
- > GRN/service completion certificate
- > Invoice
- > Payment and settlement